

TEMASEK POLYTECHNIC
SCHOOL OF INFORMATICS & IT
DIPLOMA IN CYBERSECURITY & DIGITAL FORENSICS
AY 2026/2027
PROJECT PROPOSAL
(April Semester)

ENTERPRISE NETWORKING (CCF2C06)
SUBJECT LEVEL: 2

1. This project is an individual work.
2. This assessment is worth 40% of the subject total marks. The tasks include the following:
 - a. Design IP address scheme for a given enterprise network based on assigned IP address parameters
 - b. In a proof of concept, set up DNS server on a Windows Server provisioned on AWS EC2 cloud platform via your AWS Learner Lab.
 - c. Present your and defend your proposal during Viva/Q&A.
3. The deliverables are as follows:
 - a. IP Addressing
 - A network diagram with clearly labeled subnets
 - An IP allocation table showing:

Network Segment	Subnet Address	CIDR	Usable Range	Gateway	Number of Hosts
Eg: HQ_Admin	X.X.X.X	/24	X.X.X.1 - X.X.X.254	X.X.X.254	254

- Explain how the first set of IP address can be obtained if you were to manually calculate.

- For completeness, provide the information of unused IP addresses which are for future use.
- b. Set up of DNS Server on AWS EC2 instance
- Screenshots of service configurations
 - Step-by-step implementation guide
 - Validation tests
4. The breakdown of marks (40%) award as follows:
- a. Proposal: IP Addressing scheme (8%).
 - b. Set up of DNS Server on AWS EC2 instance and demonstrate functionality. (8%)
 - c. Report (8%) – Include non-plagiarism, use of Gen AI declaration, Appendix of citations if necessary.
 - d. Viva Q&A (16%) to defend your project proposal.

Deadlines

Report Submission: 25 May 2026 9 AM.

Demonstration - During workshop in Week 6/7 (Tutor to inform you)

Please take note of the following penalties with regard to late submissions:

- Late within 1 day: 10% deduction from absolute mark given for the assignment e.g 75 marks (100 marks max) becomes 65 marks (deduct 10% of 100 marks)
- Late > 1 day and up to 2 days: 20% deduction from absolute mark
- Late > 2 days: 0 marks awarded

Project Description

You are a newly hired network engineer for a networking consultancy company, Enterprise Network Solutions Pte Ltd.

Task: IP addressing

You are given a network topology to plan out an IPv4 addressing VLSM scheme based on the given internal private IP address in Appendix A. Note: You are to use the parameters based on our class register serial number. For example, if your class register is Number 1, then you should use the following row of parameters: Packet Tracer Topology X and 10.1.0.0/16.

Class Register Number	Packet Tracer Topology Assigned	Internal Private IP address to use for subnetting
1	Topology X	10.1.0.0/16

The host number requirements is given in Appendix B.1 and Appendix B.2 for the Topology X and Topology Y respectively.

In your report, include a network diagram with clear labels. **ALL** subnets (including point-to-point networks) must be included in the IP allocation table with the following information:

Network Segment	Subnet Address	CIDR	Usable Range	Gateway	Number of Hosts
Eg: HQ_Admin	X.X.X.X	/24	X.X.X.1 - X.X.X.254	X.X.X.254	254

Explain how the first set of IP addresses can be obtained if you were to manually calculate. Include in the table, any unused IP addresses which are for future use.

NOTE: You are **NOT** required to configure the packet tracer file.

Task: Set up DNS service on a Windows Server AWS EC2 instance.

In current times, the enterprise networks have been extended to CLOUD computing platforms such as the Microsoft Azure Cloud, Google Cloud, AWS and so on.

As a proof of concept, you have been tasked to do the following on AWS Cloud:

1. set up DNS service on a Windows Server residing on AWS EC2 instance
2. Add an A record onto the DNS server such as www.mytest.com pointing to (8.8.8.8).
3. Set up your laptop to use the DNS service and ping to www.mytest.com.
4. Successful pings means your DNS service is working correctly.
5. The steps to set up the DNS must be captured accurately in the report.

All the BEST!!

IP Address Proposal (8 Marks)

Grade	Criteria
A (Excellent)	Perfect VLSM design using correct serial number parameters. IP table is complete, includes all segments (including point-to-point /30s), and clearly identifies future-use addresses. Manual calculation explanation is flawless.
B (Good)	Correct VLSM application for most networks. Table is largely complete but may have minor labeling errors. Future-use addresses are mentioned. Manual calculation is logically sound.
C (Satisfactory)	Basic subnetting applied correctly. Some errors in CIDR notation or host ranges. Point-to-point networks might be missing or incorrectly sized. Manual calculation explanation is vague.
D (Marginal)	Significant errors in IP ranges or overlapping subnets. Fails to use assigned parameters from Appendix A. Incomplete IP allocation table.
F (Fail)	Failure to provide a viable IP scheme. No evidence of manual calculation or VLSM logic.

DNS Setup on AWS EC2 (8 Marks)

Grade	Criteria
A (Excellent)	Comprehensive proof of concept on AWS. Screenshots clearly show Windows Server DNS role, the specific A-record (www.mytest.com pointing to 8.8.8.8) , and successful laptop ping results.
B (Good)	Successful setup and record creation. Verification screenshots are present but may lack detail in the step-by-step implementation guide.
C (Satisfactory)	DNS service is installed, and record is created, but validation (ping test) may be inconsistent or screenshots are poorly labeled.
D (Marginal)	DNS service installed but incorrectly configured. Record does not point to the required IP or validation tests are missing.
F (Fail)	Failure to provision the EC2 instance or configure the DNS service. Functional demonstration was not achieved.

Report (8 Marks)

Grade	Criteria
A (Excellent)	Professional layout with clear network diagrams. Includes mandatory non-plagiarism, Gen AI declaration, and proper citations. Submitted on time (by 25 May 2026 9 AM).
B (Good)	Well-organized report. Diagram is present and mostly labeled. Includes necessary declarations. Minor formatting issues.
C (Satisfactory)	Report follows basic requirements. Diagram might be cluttered or lack clear segment labels. Declarations are present but citations may be incomplete.
D (Marginal)	Poorly organized. Missing clear network labels or required declarations. Significant formatting issues.
F (Fail)	Failure to include diagrams or academic integrity declarations. Late submission beyond 2 days (0 marks).

Viva/Q&A (16 Marks)

Grade	Criteria
A (Excellent)	Demonstrates deep understanding of the proposed network. Confidently defends IP choices and cloud setup. Can explain subnetting logic and DNS resolution process perfectly.
B (Good)	Good grasp of the project. Answers most questions accurately with minor hesitation in technical explanations.
C (Satisfactory)	Basic understanding. Can explain what was done but struggles with the "why" behind specific VLSM or AWS configurations.
D (Marginal)	Weak defense. Unable to explain technical choices or manual calculation logic when prompted.
F (Fail)	Unable to answer basic questions about their own project. Demonstrates lack of ownership or technical knowledge.

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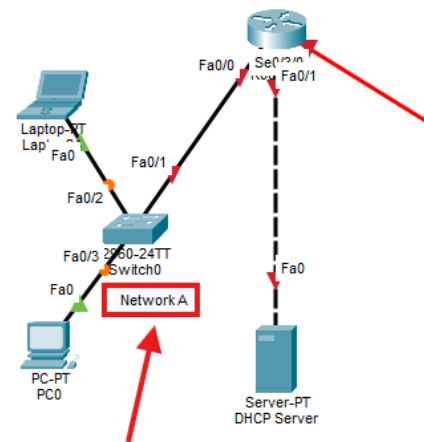
Appendix A

Class Register Number	Packet Tracer Topology Assigned	Internal Private IP address to use for subnetting
1	Topology X	10.1.0.0/16
2	Topology Y	10.2.0.0/17
3	Topology X	10.3.0.0/18
4	Topology Y	10.4.0.0/19
5	Topology X	10.5.128.0/19
6	Topology Y	172.16.0.0/16
7	Topology X	172.17.0.0/17
8	Topology Y	172.18.0.0/18
9	Topology X	172.19.0.0/19
10	Topology Y	10.1.0.0/16
11	Topology X	10.2.0.0/17
12	Topology Y	10.3.0.0/18
13	Topology X	10.4.0.0/19
14	Topology Y	172.20.128.0/19
15	Topology X	172.16.0.0/16
16	Topology Y	172.17.0.0/17
17	Topology X	172.18.0.0/18
18	Topology Y	172.19.0.0/19
19	Topology X	10.3.0.0/18
20	Topology Y	10.4.0.0/19
21	Topology X	10.4.128.0/19
22	Topology Y	172.16.0.0/16
23	Topology X	172.17.0.0/17
24	Topology Y	172.18.0.0/18
25	Topology X	172.19.0.0/19
26	Topology Y	10.1.0.0/16
27	Topology X	10.2.0.0/17
28	Topology Y	172.18.0.0/18
29	Topology X	172.19.0.0/19
30	Topology Y	10.4.0.0/19

Appendix B.1 – Network **Topology X** Host Requirements

Note: For all other point-to-point networks, assume is a 2 host network; ie use /30

Class Register Number	Number of HOSTS required on these networks									
	Network A	Network B	Network C	Network D	Network E	Network F	Network G	Network H	Network J	Network K
1	80	127	181	49	193	38	154	166	164	162
3	140	106	74	40	192	51	139	44	172	122
5	177	153	118	165	42	183	31	91	141	63
7	97	143	130	161	115	148	190	97	22	142
9	174	103	66	48	85	106	80	186	17	189
11	156	186	55	155	70	174	172	64	61	96
13	66	166	42	138	57	198	136	149	113	100
15	260	76	126	137	145	286	81	420	191	169
17	131	87	102	671	110	131	89	186	88	156
19	116	176	165	162	143	170	303	134	422	108
21	139	181	150	252	677	595	82	226	181	143
23	314	338	131	364	45	187	105	148	180	135
25	712	151	757	76	197	178	102	331	221	160
27	806	588	774	376	59	82	301	982	132	187
29	1478	111	885	49	152	86	183	172	110	98

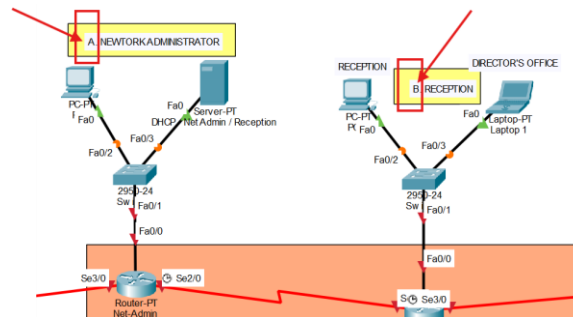


Refer to Packet Tracer for the Network Labels:

Appendix B.2 – Network **Topology Y** Host Requirements

Note: For all other point-to-point networks, assume is a 2 host network; ie use /30

Class Register Number	Number of HOSTS required on these networks														
	Network A	Network B	Network C	Network D	Network E	Network F	Network G	Network H	Network J	Network K	Network L	Network M	Network N	Network P	Network Q
2	117	21	10	119	228	89	84	189	80	63	91	113	25	109	108
4	81	115	74	77	24	99	83	97	496	155	74	74	84	261	1100
6	195	52	316	81	41	352	159	107	407	416	112	78	526	30	94
8	50	86	90	591	45	117	62	45	110	463	11	355	136	394	111
10	97	98	175	105	39	98	68	155	85	104	678	39	111	115	81
12	131	97	79	787	229	409	26	132	57	89	70	117	65	108	102
14	957	76	34	11	47	111	34	572	84	97	77	40	105	53	80
16	89	531	98	94	110	87	90	11	312	98	217	207	406	46	19
18	80	62	77	86	518	97	68	116	107	737	116	58	277	551	390
20	51	34	338	99	86	50	30	112	109	99	341	84	87	553	59
22	199	340	516	530	105	78	91	819	117	402	675	266	99	945	515
24	635	162	672	669	344	117	107	111	66	152	79	105	782	64	85
26	646	554	643	83	469	854	98	80	302	85	110	98	607	84	63
28	551	37	469	647	812	109	29	87	643	181	479	117	617	280	971
30	31	359	84	143	505	114	28	94	338	221	263	536	689	529	423



Refer to Packet Tracer for the Network Labels: